

Integral Relations among Totient Sections

Finite-level bases and bounded residue series

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ABSTRACT

The totient sections through level $e \geq 1$ in an integer base $k \geq 2$ span a rational vector space of dimension $k^e + 1$. We give a basis and determine all integral relations among the indexed sections. The proof uses the Chinese remainder theorem and Dirichlet's theorem to isolate one affine totient value at a time. The same construction gives irrationality results for bounded functions of totient residues. A final section states a sufficient exponential-sum estimate for the unresolved irrationality of $\sum_{n \geq 1} \varphi(n)2^{-n}$.

1 Introduction

We study the linear relations among the sequences $n \mapsto \varphi(k^j n + r)$, where $k \geq 2$, $j \geq 0$ and $0 \leq r < k^j$. Here φ is Euler's totient, $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\varphi(0) = 0$. At base two the identities

$$\varphi(4n) = 2\varphi(2n), \quad \varphi(4n + 2) = \varphi(2n + 1)$$

reduce the seven indexed sections through level two to five. We shall prove that these five are independent, and that in every base the corresponding scalar reductions account for all relations.

The sections form the k -kernel of the totient sequence. A sequence is k -regular when the $\mathbb{Z}$ -module generated by its k -kernel is finitely generated [1]. Coons proved that the totient sequence is not k -regular for any $k \geq 2$ [2, Thm. 3.2]. Our result specifies the rank at each finite level and gives integral coordinates for both the sections and their relations.

Theorem 1.1 (A basis through each finite level). *Let $k \geq 2$ and $e \geq 1$ be integers, write $F_{j,r}^{(k)}(n) = \varphi(k^j n + r)$, and put*

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$$V_{k,e} = \text{span}_{\mathbb{Q}} \{ F_{j,r}^{(k)} : 0 \leq j \leq e, 0 \leq r < k^j \}.$$

Then $\dim_{\mathbb{Q}} V_{k,e} = k^e + 1$, and

$$\mathcal{B}_{k,e} = \{ F_{0,0}^{(k)}, F_{1,0}^{(k)} \} \cup \{ F_{j,r}^{(k)} : 1 \leq j \leq e, 1 \leq r < k^j, k \nmid r \}$$

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is a basis. Every omitted section reduces to a basis element by an explicit scalar: $F_{j,0}^{(k)} = k^{j-1}F_{1,0}^{(k)}$ for $j \geq 1$, and if $r = k^t u$ with $t = \max\{s : k^s \mid r\} \geq 1$ and $k \nmid u$, then

$$F_{j,r}^{(k)} = C_k(t, u) F_{j-t, u}^{(k)}, \quad C_k(t, u) = k^t \prod_{\substack{p \mid k \\ p \nmid u}} \left(1 - \frac{1}{p}\right).$$

For a composite base, we retain some residues having a common factor with the base. At base six, for example, both $r = 1$ and $r = 2$ occur in the basis, although the corresponding reductions have different multipliers:

$$\varphi(36n + 6) = 2\varphi(6n + 1), \quad \varphi(36n + 12) = 4\varphi(6n + 2).$$

The factor changes because $6n + 2$ already contains the prime 2. We use the same formula when the base has repeated prime factors, so neither primality nor squarefreeness is required.

The independence of totients along pairwise nonproportional affine forms already follows from Martin's 2006 theorem on simultaneous totient inequalities [4, Thm. 1]. We give a direct proof by constructing an evaluation matrix that is nonsingular modulo a prime, using the CRT–Dirichlet separation argument in Yazdani's proof of Theorem 2, credited there to Shallit [12, pp. 652–653]. Once independence is established, the local formula for φ identifies every section omitted from the basis. Corollary 2.1 records the resulting relation module, and Section 3 works it out at base six.

Section 4 applies prime isolation to bounded functions of totient residues. The series

$$S = \sum_{n \geq 1} \frac{\varphi(n)}{2^n}$$

has unbounded coefficients. Its irrationality, asked in Erdős Problem #249 [10, p. 61], remains open. The basis theorem concerns coefficient relations; it supplies no irrationality conclusion for this value. Sections 5 and 6 describe the additional tail estimate that would suffice. Complete secondary arguments are collected in *The Binary Totient Series*.

2 Proof of the basis theorem

We first reduce the omitted sections. For $n > 0$, multiplication of kn by k^{j-1} changes no prime divisor, so $\varphi(k^j n) = k^{j-1} \varphi(kn)$; at $n = 0$ both sides vanish. If $r = k^t u > 0$ with $k \nmid u$, each prime dividing k either already divides $k^{j-t}n + u$, or contributes one new factor $1 - 1/p$ upon multiplication by k^t . The product formula for φ gives $C_k(t, u)$ and proves spanning.

For independence, let $L_i(n) = a_i n + b_i$ ($1 \leq i \leq s$), where $a_i > 0$ and $a_i b_j - a_j b_i \neq 0$ for $i \neq j$. After shifting the variable we may assume that all $b_i > 0$, and we write $g_i = \gcd(a_i, b_i)$ and $A_i = L_i/g_i$. We shall isolate column i by making the primitive part $A_i(n)$ prime and forcing a suitable prime factor into every other form: the latter factors will make all the off-diagonal totients vanish modulo a fixed prime ℓ .

Choose an odd prime ℓ dividing none of the a_i , the $g_i \varphi(g_i)$ or the nonzero cross determinants. For each fixed i , Dirichlet's theorem allows us to choose distinct primes

$q_{ij} \equiv 1 \pmod{\ell}$ ($j \neq i$) outside the same finite exceptional set. The congruences

$$A_i(n) \equiv 2 \pmod{\ell}, \quad L_j(n) \equiv 0 \pmod{q_{ij}} \quad (j \neq i)$$

are individually soluble and have pairwise coprime moduli, so CRT combines them into a progression $n = n_0 + Ru$, where $R = \ell \prod_{j \neq i} q_{ij}$.

To apply Dirichlet again, we must check that the induced progression for $A_i(n)$ is reduced modulo $(a_i/g_i)R$. Primitivity gives $\gcd(A_i(n), a_i/g_i) = 1$, while the residue 2 excludes ℓ . If some q_{ij} divided $A_i(n)$, it would divide both $L_i(n)$ and $L_j(n)$ and hence their cross determinant $a_i b_j - a_j b_i$, contrary to our choice of q_{ij} . Since these account for every prime divisor of the modulus, Dirichlet supplies arbitrarily large prime values $p = A_i(n)$, from which we choose one with $p \nmid g_i$.

At the resulting point n_i ,

$$\varphi(L_i(n_i)) = \varphi(g_i)(p-1) \equiv \varphi(g_i) \not\equiv 0 \pmod{\ell}.$$

For $j \neq i$ we have $q_{ij} - 1 \mid \varphi(L_j(n_i))$, so every other entry of row i vanishes modulo ℓ . We may choose the points n_i independently and beyond any prescribed threshold; the resulting evaluation matrix is diagonal modulo ℓ , with nonzero diagonal entries, and therefore has a nonzero integer determinant. This proves rational independence of the sequences $\varphi(L_i(n))$ after deletion of any finite prefix. In particular, the argument requires only one prime value in each row.

The two zero-residue forms in $\mathcal{B}_{k,e}$ are proportional, so we now treat them separately. Restrict a proposed relation to $n = km + 1$, where $\varphi(kn) = \varphi(k)\varphi(n)$. The positive-residue sections become the forms

$$k^{j+1}m + k^j + r \quad (k \nmid r),$$

together with $km + 1$. These are pairwise nonproportional: equality of two intercept-to-slope ratios would give $r/k^j = r'/k^{j'}$, which for $j' > j$ forces $k \mid r'$. No positive-residue form is proportional to $km + 1$. Applying affine independence now eliminates every positive-residue coefficient and gives $A + B\varphi(k) = 0$, where A, B are the coefficients of $\varphi(n), \varphi(kn)$. We obtain the second equation $A + Bk = 0$ by evaluating at $n = k$, and $\varphi(k) < k$ then forces both coefficients to vanish. Finally,

$$|\mathcal{B}_{k,e}| = 2 + \sum_{j=1}^e (k^j - k^{j-1}) = k^e + 1,$$

which completes the proof of Theorem 1.1. For independence on a tail, use a sufficiently large power of k in the last evaluation.

Corollary 2.1 (Integral coordinates and all integral relations). *Let $k \geq 2$ and $e \geq 1$. The retained family is a $\mathbb{Z}$ -basis of the module generated by the sections through level e . Index the sections by their level and residue, retaining distinct indices even when they define equal sequences. For each omitted index i , write the scalar reduction as $F_i = a_i F_{j(i)}$, where $j(i)$ is retained and a_i is a nonnegative integer. In the free abelian group with one generator E_i for each of these indices, the vectors*

$$R_i = E_i - a_i E_{j(i)} \quad (i \text{ omitted})$$

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form a $\mathbb{Z}$ -basis of the kernel of evaluation $E_i \mapsto F_i$. In particular, every integral relation has a unique integral expression in these elementary relations, and their rank is $\sum_{j=1}^{e-1} k^j$.

Proof. Since $\prod_{p|k, p \nmid u} p$ divides k , every scalar $C_k(t, u)$ is an integer. The retained family therefore spans the section module over $\mathbb{Z}$, and its rational independence makes it a $\mathbb{Z}$ -basis.

Let $x = \sum_i x_i E_i$ evaluate to zero. If we subtract $\sum_{i \text{ omitted}} x_i R_i$, the remaining vector is supported on retained indices and still evaluates to zero, so it vanishes by their independence. Conversely, we recover c_i from the coefficient at omitted index i in any combination $\sum_h c_h R_h$, proving both independence of the R_i and uniqueness of the expansion. Counting omitted indices gives $(1 + k + \dots + k^e) - (k^e + 1) = \sum_{j=1}^{e-1} k^j$. $\square$

Note that the section module can be smaller than the group of integer-valued sequences in its rational span. For $k = 2$ and $e \geq 2$, for example, $\varphi(4n + 3)/2$ is integer-valued but has coefficient $1/2$ in the retained basis, so it does not belong to the module. Thus saturation in all integer-valued sequences fails.

Corollary 2.2 (Dyadic basis). *The family*

$$\{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1, 0 < r < 2^j, r \text{ odd}\}, \quad \varphi_{j,r}(n) = \varphi(2^j n + r),$$

is a basis for the rational span of the full dyadic kernel. Every rational relation is generated by the scalar reductions. The complete level-zero truncation has dimension one.

Proof. Each finite subset of this family is contained in some $\mathcal{B}_{2,e}$. Theorem 1.1 gives independence, and its reductions give spanning. At level zero the sole section $\varphi(n)$ is nonzero, since $\varphi(1) = 1$. $\square$

The affine argument also permits periodic coefficients by restriction to their common residue classes.

Corollary 2.3 (Periodic coefficients). *Let $L_1, \dots, L_s$ be pairwise nonproportional affine forms with integer coefficients and positive slopes. If $w_1, \dots, w_s$ are rational-valued periodic sequences, then*

$$\sum_i w_i(n) \varphi(L_i(n)) = 0 \quad \text{for all sufficiently large } n \implies w_i(n) = 0 \quad \text{for every } i, n.$$

Proof. Choose a common period Q and restrict to $n = r + Qt$, where the coefficients become the constants $w_i(r)$ and the cross determinants become $Q(a_i b_j - a_j b_i) \neq 0$. We may shift t beyond the given threshold and apply affine independence to obtain $w_i(r) = 0$ for every i . Repeating this for each residue r and using periodicity proves the conclusion at every index. $\square$

Proportional forms are excluded for a reason: the identity $\varphi(4n + 2) = \varphi(2n + 1)$ would otherwise contradict the conclusion. Martin's theorem gives the same corollary on each progression [4, Thm. 1].

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3 An equality test at base six

Consider the 43 indexed sections through level two in base six. The 37 retained sections are $F_{0,0}, F_{1,0}$, the five $F_{1,r}$ ($1 \leq r \leq 5$), and the 30 sections $F_{2,r}$ with $1 \leq r < 36$ and $6 \nmid r$. The omitted sections satisfy

$$\begin{aligned} F_{2,0} &= 6F_{1,0}, & F_{2,6} &= 2F_{1,1}, & F_{2,12} &= 4F_{1,2}, \\ F_{2,18} &= 3F_{1,3}, & F_{2,24} &= 4F_{1,4}, & F_{2,30} &= 2F_{1,5}. \end{aligned}$$

Thus an integral combination of the 43 sections vanishes precisely when these six substitutions leave all 37 retained coefficients zero. If it vanishes, its six original omitted coefficients are its unique coordinates in the relation basis of Corollary 2.1. At base two and level two the corresponding relation basis is simply $E_{2,0} - 2E_{1,0}$, $E_{2,2} - E_{1,1}$.

We can also decide equality by finitely many evaluations. Order the retained columns as above, with residues increasing within each level, and evaluate at $n = 0, \dots, 36$ to obtain a 37×37 matrix. Its determinant is 17 modulo 101, so it is non-singular over $\mathbb{Q}$ and detects every nonzero retained combination at one of these inputs. The script `scripts/totient_kernel_normal_form.py` calculates the coordinates and the first such input using integer arithmetic. The equality test uses the integer evaluations. Reducing the proposed combination modulo 101 would miss $101F_{2,2}$, for example, although its value at $n = 0$ is 101.

4 Bounded functions of totient residues

Prime isolation also distinguishes rational from irrational values for bounded functions of $\varphi(n)$ modulo a fixed integer. We use the least nonnegative representative for a residue in an ordinary sum.

Theorem 4.1 (Residue series and dyadic observables). *For every $m \geq 3$,*

$$\sum_{n \geq 1} \frac{\varphi(n) \bmod m}{2^n} \notin \mathbb{Q}.$$

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For $k \geq 1$ and $f : \mathbb{Z}/2^k\mathbb{Z} \rightarrow \mathbb{Q}$, the series $\sum_{n \geq 1} f(\varphi(n) \bmod 2^k) 2^{-n}$ is rational exactly when f is constant on the even residue classes. If that constant is c , its value is $3f(1)/4 + c/4$.

Modulo two the sum is $3/4$, since $\varphi(n)$ is even for $n \geq 3$. More generally only $f(1)$ and the even residue classes occur in the second assertion. Its proof uses the following elementary tail argument.

Lemma 4.2 (An isolated nonzero coefficient between long zero blocks). *Let $a_n \in \mathbb{Z}$ satisfy $|a_n| \leq C$. Suppose that for arbitrarily large L there is $N > L$ such that $a_N \neq 0$ and $a_{N+t} = 0$ for $0 < |t| \leq L$. Then $\sum_{n \geq 1} a_n 2^{-n}$ is irrational.*

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Proof. Suppose that the sum has denominator $q \geq 1$, so that every scaled tail $T_j = \sum_{t \geq 1} a_{j+t} 2^{-t}$ belongs to $q^{-1}\mathbb{Z}$. We choose an isolated coefficient with $C2^{-L} < 1/q$. The zero block to its right gives $|T_N| \leq C2^{-L} < 1/q$ and therefore $T_N = 0$, whereas the block to its left then gives $T_{N-L-1} = a_N 2^{-L-1}$, a nonzero element of $q^{-1}\mathbb{Z}$ of absolute value less than $1/q$. This is a contradiction. $\square$

For example, the indicator of the squares satisfies the lemma. An eventually periodic sequence with infinitely many nonzero terms does not have isolated coefficients at arbitrarily large distances. The uniform bound C is essential to this argument.

Proof of Theorem 4.1. Fix a unit $s \pmod{m}$ and a positive integer L . Choose distinct primes $q_t \equiv 1 \pmod{m}$, all exceeding L , for $0 < |t| \leq L$. The congruences

$$p \equiv s \pmod{m}, \quad p \equiv -t \pmod{q_t} \quad (0 < |t| \leq L)$$

define a reduced progression, since s is a unit modulo m and $0 < |t| < q_t$. CRT and Dirichlet's theorem therefore allow us to choose an arbitrarily large prime $p > L + 1$ on this progression, with $m \mid \varphi(p + t)$ at every neighbour and $\varphi(p) \equiv s - 1 \pmod{m}$ at the centre.

Subtract $f(0)$ and multiply by a common denominator of the values of f . If $f(s - 1) \neq f(0)$, the resulting bounded integer sequence has an isolated nonzero coefficient for every L , so the lemma applies. For the least-residue map we take $s = -1$ and obtain the nonzero residue $m - 2$ when $m \geq 3$. When $m = 2^k$, each even residue r is of the form $s - 1$ with s a unit, so nonconstancy on the even classes gives an irrational sum by the same argument. Conversely, if f is constant there, all terms after $n = 2$ equal c and we obtain $3f(1)/4 + c/4$. $\square$

The first assertion is also an application of Erdős's sparse-series criterion [6, Lem. 1, p. 213], and of Kaneko, Suzuki and Tachiya [7, Thm. 3]. The nonzero residues occur on a set of density zero: an integer in that set has no prime divisor $p \equiv 1 \pmod{m}$. Wong gave the case in which the base equals the modulus [5]. The isolated-coefficient argument above also treats signed values of $f - f(0)$, for which those nonnegative sparse-series criteria need an additional spacing hypothesis. The [recorded formal extension](#) treats every integer base $t \geq 2$ and proves rational linear independence of 1 and $\sum_n (\varphi(n) \bmod m) t^{-dn}$, $d \geq 1$.

5 The binary totient series

For the unreduced totient, $0 \leq \varphi(n) \leq n$ gives convergence of S , but no fixed coefficient bound for Lemma 4.2. In fact, forcing $2^L \mid \varphi(N + L)$ with $N + L > 1$ gives $2^L \leq N + L - 1$. The resulting tail envelope $(N + L + 2)2^{-L} > 1$ cannot exclude a positive rational denominator. We instead consider differences of scaled tails.

Put $R_N = \sum_{j \geq 1} \varphi(N + j)2^{-j}$ and $\Delta_h(N) = R_{N+h} - R_N$. Subtracting the finite-prefix identities for $2^{N+h}S$ and $2^N S$ gives

$$\Delta_h(N) \equiv 2^N(2^h - 1)S \pmod{\mathbb{Z}}. \quad (1)$$

If $S = a/(2^c v)$, with v odd, then $\Delta_h(N)$ is integral for $N \geq c$ and $h = \varphi(v)$. To disprove rationality it therefore suffices to find a nonintegral difference beyond every threshold for each positive h .

For $h, N, L \in \mathbb{N}$, define the integer

$$D_{h,N,L} = \sum_{j=0}^{L-1} (\varphi(N + h + 1 + j) - \varphi(N + 1 + j))2^{L-1-j}, \quad B = N + h + L + 2.$$

Since $0 \leq R_M \leq M + 2$, we have

$$2^L \Delta_h(N) - D_{h,N,L} = R_{N+h+L} - R_{N+L}, \quad |2^L \Delta_h(N) - D_{h,N,L}| \leq B. \quad (2)$$

The difference is bounded by the larger tail bound. Hence the finite inequality

$$B < D_{h,N,L} \bmod 2^L < 2^L - B \quad (3)$$

forces $\Delta_h(N) \notin \mathbb{Z}$. Conversely, if its distance to $\mathbb{Z}$ is positive, choose L so that $2^L \|\Delta_h(N)\|_{\mathbb{R}/\mathbb{Z}} > 2B$; the same error estimate gives (3). Thus allowing an unrestricted depth loses no nonintegral difference. For example, $D_{1,12,16} = -143140$ has residue 53468 modulo 2^{16} , and $B = 31$, so it excludes denominators dividing 2^{12} .

5.1 Equivalent residue conditions

We write $K(h, N, L)$ for (3).

Theorem 5.1 (Propagation of one nonintegral tail difference). *Fix $d \geq 1$ and $N \geq 0$. If $\Delta_d(N) \notin \mathbb{Z}$, then every sufficiently late pair $\{t, t+1\}$ contains an m such that $K(md, N, md)$ holds. Consequently*

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$$\exists t \geq 1 : K(td, N, td) \iff \Delta_d(N) \notin \mathbb{Z}.$$

Requiring this for every $d \geq 1, N \geq 0$ is equivalent to $S \notin \mathbb{Q}$.

The proof uses the recurrence $\Delta_{(t+1)d}(N) \equiv 2^d \Delta_{td}(N) + \Delta_d(N) \pmod{\mathbb{Z}}$; it is given in [Appendix A of the long record](#). A single pair d, N need not imply irrationality; the final assertion requires every positive d and every $N \geq 0$.

There is a second equivalent test for each hypothetical denominator $2^c v$. For a positive multiple H of $\varphi(v)$, set $M = (2^H - 1)/v$ and $B_{H,c} = \sum_{j=0}^{H-1} \varphi(c+1+j)2^{H-1-j}$.

Equivalent formulation 5.2 (A residue test modulo a divisor of $2^H - 1$). The series S is irrational if and only if, for every $c \geq 0$ and odd $v \geq 1$, some positive multiple H of $\varphi(v)$ satisfies

$$c + H + 1 < (-B_{H,c}) \bmod M < M - (c + H + 1). \quad (4)$$

The [proof and residue recurrence](#) use $B_{H,c} = M(vR_c) - \Delta_H(c)$ and the linear tail bound. Neither test provides these universal witnesses. A separate [Farey-interval calculation](#) at depth 240 excludes $S = a/q$ for $0 < q \leq Q_0 \approx 7.96 \times 10^{34}$.

5.2 Comparison sequences

Large ranks alone cannot settle the series question. There is an integer sequence c with $0 \leq c(n) \leq n$, unchanged odd totients and $|c(n) - \varphi(n)| \leq 2$ at even indices, for which $\sum c(n)2^{-n} = 5/4$. Its integral carry still has dyadic rank at least $2^e - 1$ through every positive level. The [construction and its limitations](#) explain why it fails the exact prime-dilation laws of φ . The same section retains the cofinal LCM-diagonal criterion, its endpoint sign conditions and the limits of finite integer combinations of shifts.

A more flexible [signed interpolation theorem](#) fills an interval about $\sum \varphi(n)\beta^{-n}$, for each integer $\beta \geq 2$, while preserving a prescribed prefix, all odd coefficients and errors bounded by any positive $f(n) \rightarrow \infty$. Its even support has upper Banach density

zero. Every fixed progression has an entire Dirichlet defect. At each root of unity the generating-function defect has smooth radial data, with the prescribed finite jets zero. The theorem is an ordinary proof; it preserves Dirichlet singularities, not regular parts or zeros, and finite jets do not specify an entire radial germ. In particular it does not prove cancellation for the unchanged totient.

Rational approximation imposes a different requirement. The identity

$$\Theta_2 := \sum_{d \geq 1} \frac{\mu(d)}{(2^d - 1)^2} = S - \frac{1}{2}$$

was posted by Amiram Eldar, together with its coprimality interpretation, on 15 March 2026 [8], and by Steve Fan on 16 May 2026 [9]. Consider the rational quotients

$$Q(e, Y) = \frac{(\sum_{d \leq Y} \mu(d) / (2^d - 1)^{e+2})^2}{\sum_{d \leq Y} \mu(d) / (2^d - 1)^{2e+2}}.$$

Theorem 5.3 (A lower bound for the rank-one quotients). *For $e \geq 1$ and $Y \geq 4$, the denominator of $Q(e, Y)$ is positive, and the unique minimiser is $(e, Y) = (1, 5)$. Every admissible quotient and every nonempty finite positive weighted average of such quotients satisfies*

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$$Q - \Theta_2 > \frac{21}{320}.$$

The uniform bound $Q - \Theta_2 > 1/15$ is false already at $(1, 5)$.

The [proof in the long record](#) bounds the infinite tails geometrically and compares seven-term rational prefixes. This concerns averages taken after forming the quotients; signed combinations, coupling before division and higher-rank kernels are separate questions. A family of rational approximants p_j/q_j in lowest terms would instead need $0 < q_j|\Theta_2 - p_j/q_j| \rightarrow 0$. Mere convergence does not give it.

6 An exponential-sum question

For every $h \geq 1$ and every X_0 , suppose that some integers $X \geq \max(X_0, 1)$ and $L \geq h$ satisfy

$$16(2X + h + L + 2) \leq 2^L, \quad \operatorname{Re} \sum_{N=X}^{2X-1} \exp(2\pi i D_{h,N,L}/2^L) \leq \frac{9}{10}X. \quad (5)$$

These inequalities would prove irrationality. Indeed, if every residue failed (3), each phase would be within $1/16$ of an integer, so every real part would be at least $\cos(\pi/8) > 9/10$. A certificate must therefore occur on the block, and (1) excludes every rational denominator by choosing its period and then a sufficiently late block.

The following decomposition isolates the estimates still needed. Fix $h, s \geq 1, L \geq s+h, X \geq 1$ and $0 < \eta < 1$; put $t = L - s + 1$ and $E_N = \exp(2\pi i D_{h,N,L}/2^L)$. For $X \leq N < 2X$, let p_N be the largest prime factor of $N + t$ and $m_N = (N + t)/p_N$. Define

$$\begin{aligned} A &= \{N \in [X, 2X) : m_N \leq \lfloor \sqrt{X}/2 \rfloor, p_N > 2\lfloor \sqrt{X} \rfloor\}, \\ \mathcal{G} &= \{N \in A : \varphi(m_N) \geq \eta m_N\}. \end{aligned}$$

All indices are integers. On $\mathcal{A}$ we have $p_N > m_N$, so $\varphi(N+t) = \varphi(m_N)(p_N-1)$. Its coefficient in the phase is $(2^h-1)/2^t$. Write

$$z_N = \exp\left(\frac{2\pi i(2^h-1)\varphi(m_N)(p_N-1)}{2^t}\right), \quad w_N = E_N/z_N,$$

and let $\bar{z}_m$ be the average over the indices in $\mathcal{A}$ with cofactor m , with value zero for an empty set. The four proposed bounds are

$$\begin{aligned} \operatorname{Re} \sum_{N \in \mathcal{G}} w_N (z_N - \bar{z}_{m_N}) &\leq \frac{14}{25}X, \\ \left| \sum_{N \in \mathcal{G}} w_N \bar{z}_{m_N} \right| &\leq \frac{1}{100}X, \\ \left| \sum_{N \in \mathcal{A} \setminus \mathcal{G}} E_N \right| &\leq \frac{1}{100}X, \\ \left| \sum_{\substack{X \leq N < 2X \\ N \notin \mathcal{A}}} E_N \right| &\leq \frac{8}{25}X. \end{aligned} \tag{6}$$

Before taking real parts and absolute values, the four sums add to $\sum E_N$, and hence their bounds imply the real-part estimate $9X/10$ in (5). We have only added and subtracted the group means, without any independence assumption. The resulting bound concerns the real part, and does not give a $21X/25$ bound for the absolute value of the full sum.

Problem 6.1 (Four estimates for the first-harmonic sum). For each $h \geq 1$, do there exist $s \geq 1$ and $0 < \eta < 1$ such that, for every X_0 , there are X, L with

$$\max(X_0, 1) \leq X, \quad h \leq L - s, \quad 16(2X + h + L + 2) \leq 2^L,$$

for which those four bounds hold?

All four estimates must hold on the same block with the same parameters. For each fixed h, s , take the least admissible L satisfying the room inequality and $L \geq s + h$. The [smooth-number and small-cofactor estimates](#) give the last two bounds for all sufficiently large X when $0 < \eta < 1/(300 \log 4)$. The elementary excluded-cofactor estimate is $(3\eta \log 4 + o(1))X$; it needs no prime number theorem. The group-mean and weighted mean-subtracted estimates remain open. No optimality is asserted for the sufficient margins in (6).

Alternatively, at $s = 26$, $\eta = 1/1000$ and minimal depth one has $|\mathcal{G}| > 67X/100$ for large X . The single open estimate $\operatorname{Re} \sum_{N \in \mathcal{G}} E_N \leq 603X/1000$, for each h on arbitrarily large blocks, would suffice by the same argument. The [rational comparison sequences](#) force larger first harmonics at all sufficiently large admissible scales; this describes their difference from the required behaviour of φ .

Balasubramanian, Giri and Srivastav give correlation estimates applicable to $\varphi(n)/n$, uniformly for $|h| \leq x/2$ [11, Thm. 2.2]. Here the modulus grows with L , and the residual weight depends on other totient values in the window. That correlation theorem supplies neither of the two remaining estimates. Establishing them for the actual totient is the unresolved arithmetic step.

A Regularity and formal sources

Coons's proof of nonregularity counts the poles of $\zeta(s-1)/\zeta(s)$ [2, Cor. 3.1, Thm. 3.2]. For an integer-valued sequence, finite dimension of the rational kernel span is equivalent to finite generation of its integral kernel module: choose finitely many evaluations that embed the span in $\mathbb{Q}^d$; the module embeds in $\mathbb{Z}^d$, whose subgroups are finitely generated (compare [3, App. A]). Theorem 1.1 recovers this nonregularity and gives the ranks of its finite truncations. For an account of the earlier separation method, see Allouche, Shallit and Yassawi [13, Thm. 3, Ex. 4, Rem. 7].

Martin's theorem also proves the affine independence directly. Put a nonzero coefficient first in a proposed relation and choose the successive totient ratios larger than $\max(1, \sum_{i>1} |c_i|/|c_1|)$. On the positive-lower-density set provided by his theorem, the first term then dominates the sum of the others in absolute value [4, Thm. 1].

These facts have a functional consequence. A k -Mahler equation has the form $\sum_{i=0}^d P_i(z)F(z^{k^i}) = 0$, with $P_i \in \mathbb{Q}[z]$ and $P_0 \neq 0$. The height theorem of Adamczewski, Bell and Smertnig implies that a k -Mahler series with integer coefficients of polynomial growth is k -regular [16, Thm. 1.2(b)]. Since $\varphi(n) \leq n$, its logarithmic height is at most $\log n$, so $F_\varphi(z) = \sum_{n \geq 1} \varphi(n)z^n$ has no such equation. Bell and Smertnig's multiplicative classification gives the same conclusion [14, Thm. 1.3]. Bell, Bruin and Coons treat related algebraic and D -finite series, the latter satisfying a linear differential equation with polynomial coefficients [15, Thms. 1.5–1.6]. These statements concern the function, and do not determine $F_\varphi(1/2)$.

The section operators $T_i g(n) = g(kn + i)$ satisfy

$$T_i F_{j,r} = F_{j+1, k^j i + r}, \quad V_{k,e+1} = V_{k,e} + \sum_{i=0}^{k-1} T_i V_{k,e}.$$

To obtain the second identity, write every residue at level $j+1$ as $k^j i + r$. For a positive retained residue, the resulting section is again retained. For the two zero-residue generators, use $T_i F_{0,0} = F_{1,i}$ and $T_i F_{1,0} = F_{2,ki}$ followed by the zero-residue or scalar reduction. We thus obtain a positive integer multiple of a basis element in every case, although the increasing dimensions prevent any finite truncation from being invariant. The growth formula using the joint spectral radius for finite-dimensional regular sequences [3, Thm. 1, Prop. 4, Cor. 7] therefore does not apply to φ .

Verification. *Formal proofs.* A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again

from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text. The [declaration guide](#) preserves the individual formal-source links for the short paper, including the three results whose full proofs appear in Appendix A of the long record. The displayed arguments and the signed interpolation proof are to be read as ordinary mathematical proofs. Recorded Lean and Comparator checks refer to their exact declarations and source versions; this revision makes no claim of a new formal run or independent review.

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